

 Discrete textual token

 Bitwise discrete visual token

Text Decoder

Visual Decoder

Unified Auto-Regressive Model

Unified
Visual Tokenizer

Text Tokenizer

Text Tokenizer

Unified
Visual Tokenizer

Parallel
bitwise prediction

Bitwise d^{BSQ}

1	0	0	1	...	1	0	1	1
1	1	1	1	...	0	1	0	1
0	1	0	0	...	0	0	1	1
:	:	:	:	...	:	:	:	:
1	0	1	0	...	0	1	1	1
1	0	1	0	...	1	1	0	1
0	0	0	0	...	1	0	0	0
1	1	1	1	...	1	1	1	1

Parallel group $g = n^{level} \times n^{spatial}$